Year 10 Higher

REVISION PACK

Shape and Space 1

Name: _____

Class: _____

Shape and Space 1 · Angles (Lessons 5.1, 5.3)

Worked example: the exterior angle of a triangle

In triangle ABC, the side BC is extended to D. Angle $BAC = (x + 10)^\circ$, angle $ABC = (2x + 20)^\circ$ and angle $ACD = (4x - 5)^\circ$. Work out the size of angle ACD.

Step 1	Exterior = two opposite angles	$x + 10 + 2x + 20 = 4x - 5$
Step 2	Collect like terms	$3x + 30 = 4x - 5$
Step 3	Subtract $3x$, add 5	$x = 35$
Step 4	Angle ACD	$4 \times 35 - 5 = 135^\circ$

Practice: the exterior angle of a triangle

1.1

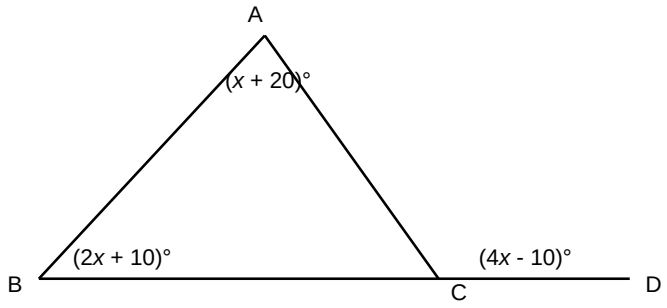


Diagram **NOT** accurately drawn

The diagram shows triangle ABC with the side BC extended to D. Work out the size of angle ACD.

1.2

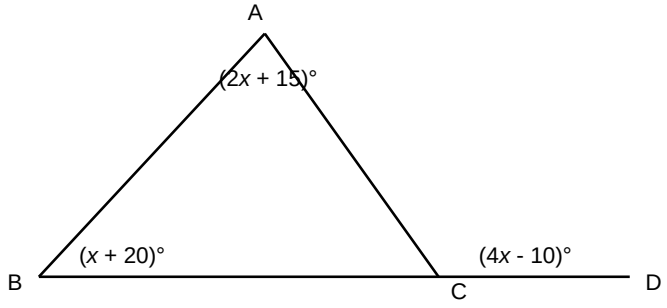


Diagram **NOT** accurately drawn

The diagram shows triangle ABC with the side BC extended to D. Work out the size of angle ACD.

1.3

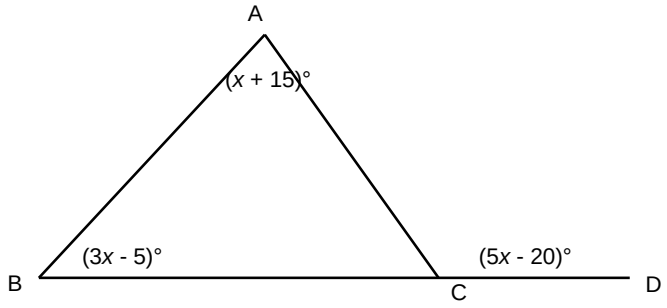


Diagram **NOT** accurately drawn

The diagram shows triangle ABC with the side BC extended to D. Work out the size of angle ACD.

Practice: the exterior angle of a triangle (continued)

1.4

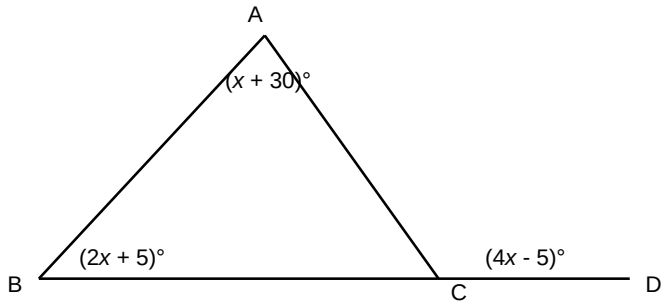


Diagram **NOT** accurately drawn

The diagram shows triangle ABC with the side BC extended to D.
Work out the size of angle ACD.

1.5

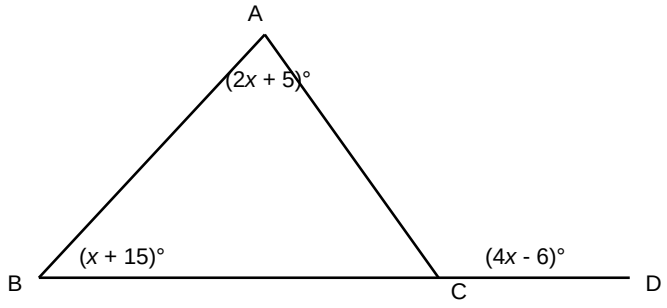


Diagram **NOT** accurately drawn

The diagram shows triangle ABC with the side BC extended to D.
Work out the size of angle ACD.

1.6

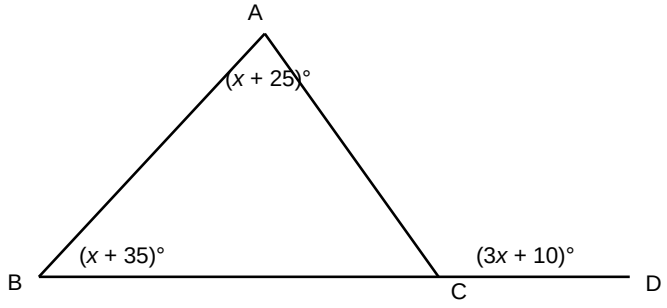


Diagram **NOT** accurately drawn

The diagram shows triangle ABC with the side BC extended to D.
Work out the size of angle ACD.

Shape and Space 1 · Angles (Lessons 5.1, 5.3)

Worked example: angles in a parallelogram

$ABCD$ is a parallelogram. Angle $A = (2x + 20)^\circ$ and angle $D = (3x + 10)^\circ$. Work out the size of the smallest angle.

Step 1	Co-interior angles add to 180°	$2x + 20 + 3x + 10 = 180$
Step 2	Collect like terms	$5x + 30 = 180$
Step 3	Subtract 30, divide by 5	$x = 30$
Step 4	Angle A	$2 \times 30 + 20 = 80^\circ$
Step 5	Angle D	$3 \times 30 + 10 = 100^\circ$
Step 6	Answer	80°

Practice: angles in a parallelogram

1.7

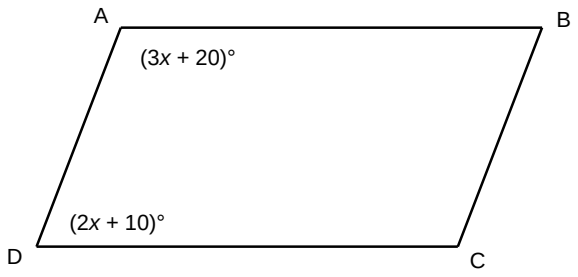


Diagram **NOT** accurately drawn

$ABCD$ is a parallelogram.
Work out the size of the smallest angle.

1.8

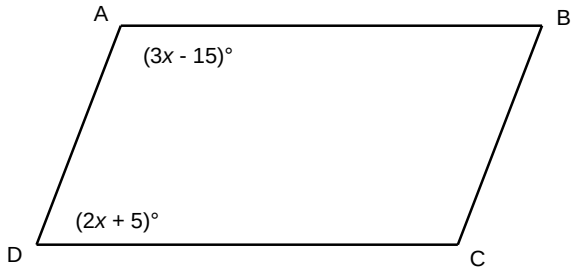


Diagram **NOT** accurately drawn

$ABCD$ is a parallelogram.
Work out the size of the smallest angle.

1.9

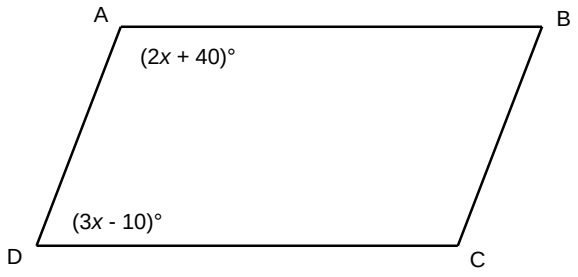


Diagram **NOT** accurately drawn

$ABCD$ is a parallelogram.
Work out the size of the smallest angle.

Practice: angles in a parallelogram (continued)

1.10

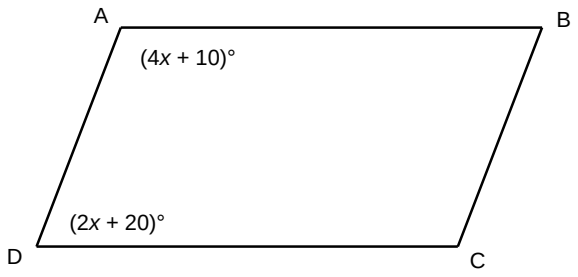


Diagram **NOT** accurately drawn

$ABCD$ is a parallelogram.
Work out the size of the smallest angle.

1.11

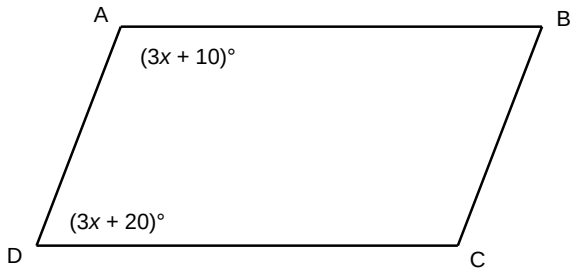


Diagram **NOT** accurately drawn

$ABCD$ is a parallelogram.
Work out the size of the smallest angle.

1.12

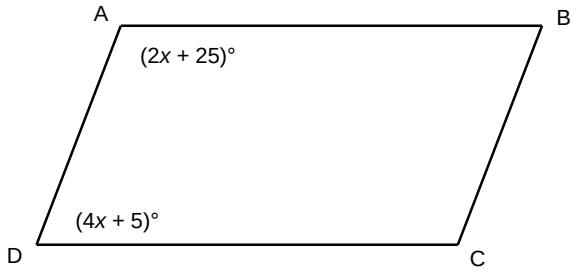


Diagram **NOT** accurately drawn

$ABCD$ is a parallelogram.
Work out the size of the smallest angle.

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.2, 5.4, 5.5)

Worked example: finding a shorter side

A right-angled triangle has hypotenuse 12.5 cm. One shorter side is 7.2 cm. Work out the length x of the other side. Give your answer correct to 3 significant figures.

Step 1	Square and subtract	$x^2 = 12.5^2 - 7.2^2 = 104.41$
Step 2	Square root	$x = \sqrt{104.41} = 10.2181\ldots$
Step 3	Answer	$10.2\ \text{cm}$

Practice: finding a shorter side

2.1

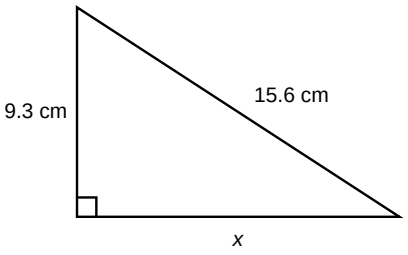


Diagram **NOT** accurately drawn

Work out the length x.
Give your answer correct to 3 significant figures.

2.2

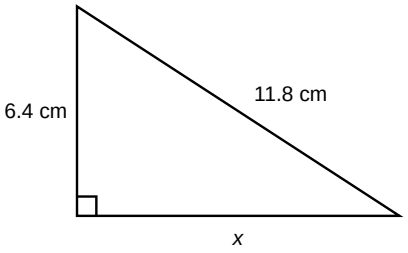


Diagram **NOT** accurately drawn

Work out the length x.
Give your answer correct to 3 significant figures.

2.3

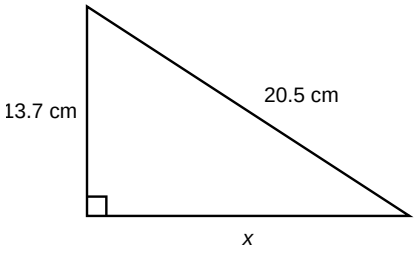


Diagram **NOT** accurately drawn

Work out the length x.
Give your answer correct to 3 significant figures.

Practice: finding a shorter side (continued)

2.4

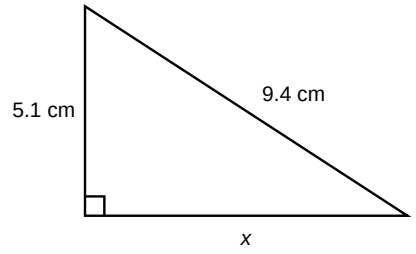


Diagram **NOT** accurately drawn

Work out the length x.
Give your answer correct to 3 significant figures.

2.5

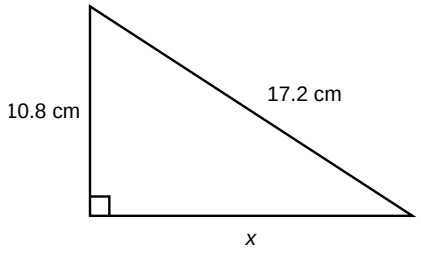


Diagram **NOT** accurately drawn

Work out the length x.
Give your answer correct to 3 significant figures.

2.6

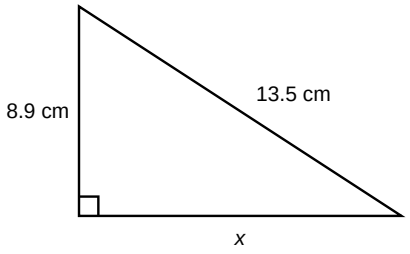


Diagram **NOT** accurately drawn

Work out the length x.
Give your answer correct to 3 significant figures.

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.2, 5.4, 5.5)

Worked example: the area of a trapezium

A trapezium has parallel sides x cm and $(x + 2)$ cm. Its height is 5 cm. The area of the trapezium is 45 cm². Work out the value of x.

Step 1	Area formula	$\frac{1}{2}(x + x + 2) \times 5 = 45$
Step 2	Multiply by 2	$(2x + 2) \times 5 = 90$
Step 3	Divide by 5	$2x + 2 = 18$
Step 4	Subtract 2, divide by 2	$x = 8$

Practice: the area of a trapezium

2.7

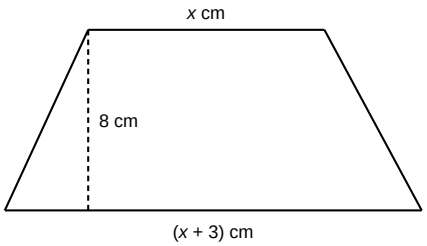


Diagram **NOT** accurately drawn

The area of the trapezium is 60 cm²
Work out the value of x.

2.8

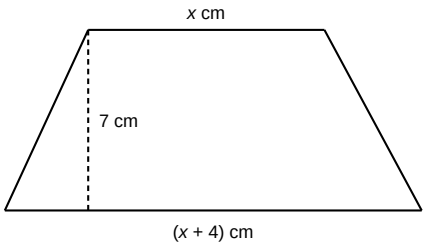


Diagram **NOT** accurately drawn

The area of the trapezium is 98 cm²
Work out the value of x.

2.9

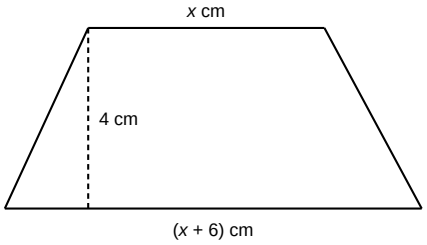


Diagram **NOT** accurately drawn

The area of the trapezium is 48 cm²
Work out the value of x.

Practice: the area of a trapezium (continued)

2.10

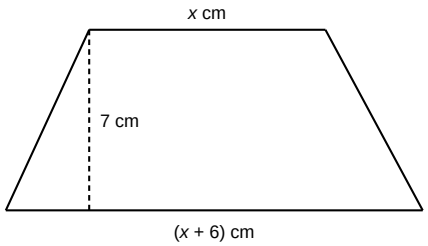


Diagram **NOT** accurately drawn

The area of the trapezium is 91 cm²
Work out the value of x.

2.11

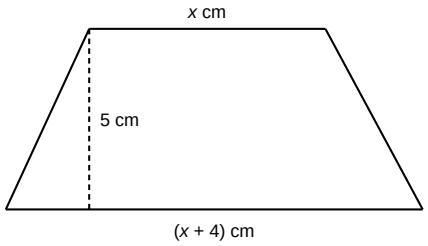


Diagram **NOT** accurately drawn

The area of the trapezium is 55 cm²
Work out the value of x.

2.12

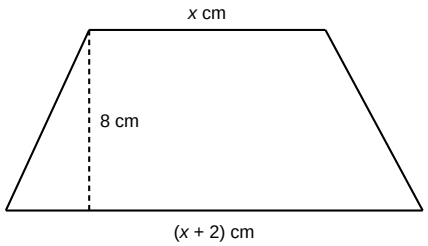


Diagram **NOT** accurately drawn

The area of the trapezium is 120 cm²
Work out the value of x.

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.2, 5.4, 5.5)

Worked example: radius from the area

A circle has area 200 cm². Work out the radius of the circle. Give your answer correct to 3 significant figures.

Step 1	$A = \pi r^2$	$\pi r^2 = 200$
Step 2	Divide by π	$r^2 = 200 \div \pi = 63.662\ldots$
Step 3	Square root	$r = 7.97885\ldots$
Step 4	Answer	$7.98\ \text{cm}$

Practice: radius from the area

2.13 A circle has area 120 cm². Work out the radius of the circle. Give your answer correct to 3 significant figures.

2.14 A circle has area 300 cm². Work out the radius of the circle. Give your answer correct to 3 significant figures.

2.15 A circle has area 85 cm². Work out the radius of the circle. Give your answer correct to 3 significant figures.

2.16 A circle has area 60 cm². Work out the radius of the circle. Give your answer correct to 3 significant figures.

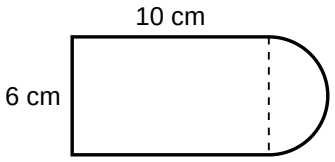
2.17 A circle has area 250 cm². Work out the radius of the circle. Give your answer correct to 3 significant figures.

2.18 A circle has area 45 cm². Work out the radius of the circle. Give your answer correct to 3 significant figures.

Shape and Space 1 · Semicircles and compound shapes (Lesson 5.5)

Worked example: a rectangle and a semicircle

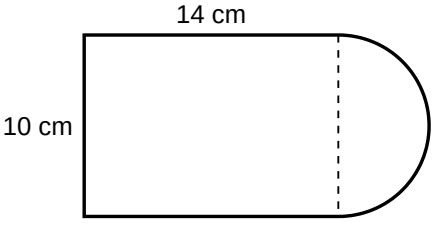
The shape is made from a rectangle and a semicircle. Work out (a) the perimeter and (b) the area. Give your answers correct to 3 significant figures.



(a)	Step 1	Add the three straight sides	$10 + 6 + 10 = 26$
	Step 2	Half the circumference, $d = 6$	$\frac{1}{2} \times \pi \times 6 = 9.424\ldots$
	Step 3	Add the straight sides	$26 + 9.424\ldots = 35.4$ cm
(b)	Step 1	Area of the rectangle	$10 \times 6 = 60$
	Step 2	Half the area of a circle, $r = 3$	$\frac{1}{2} \times \pi \times 3^2 = 14.137\ldots$
	Step 3	Add the two areas	$60 + 14.137\ldots = 74.1$ cm²

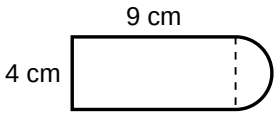
Practice: semicircles and compound shapes

3.1 The shape is made from a rectangle and a semicircle.
(a) Work out the perimeter of the shape. (b) Work out the area of the shape.
Give your answers correct to 3 significant figures.

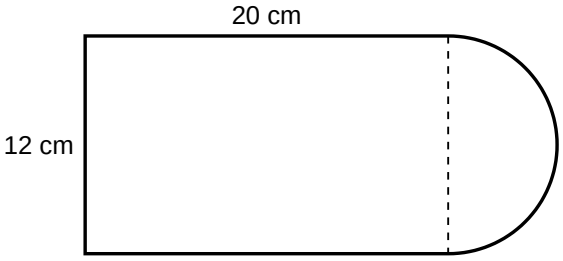


Practice: semicircles and compound shapes (continued)

3.2 The shape is made from a rectangle and a semicircle.
(a) Work out the perimeter of the shape. (b) Work out the area of the shape.
Give your answers correct to 3 significant figures.

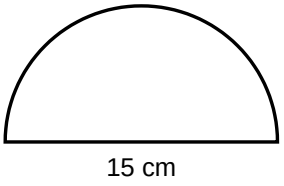


3.3 The shape is made from a rectangle and a semicircle.
(a) Work out the perimeter of the shape. (b) Work out the area of the shape.
Give your answers correct to 3 significant figures.

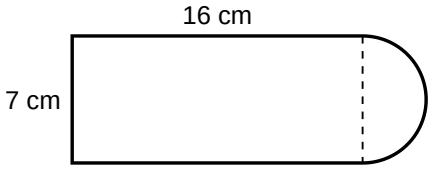


Practice: semicircles and compound shapes (continued)

3.4 The shape is a semicircle.
(a) Work out the perimeter of the shape. (b) Work out the area of the shape.
Give your answers correct to 3 significant figures.

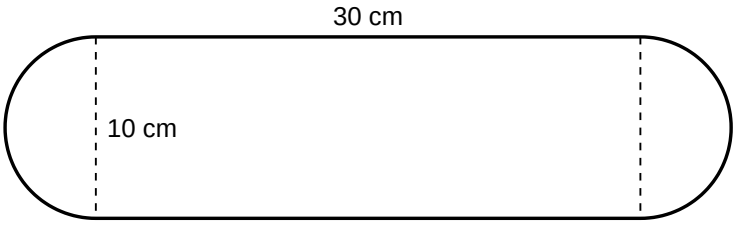


3.5 The shape is made from a rectangle and a semicircle.
(a) Work out the perimeter of the shape. (b) Work out the area of the shape.
Give your answers correct to 3 significant figures.



Practice: semicircles and compound shapes (continued)

3.6 A running track is made from a rectangle and two semicircles.
(a) Work out the perimeter of the shape. (b) Work out the area of the shape.
Give your answers correct to 3 significant figures.



Worked example: angles in a quadrilateral

The angles of a quadrilateral are $(x + 10)^\circ$, $2x^\circ$, $(3x - 20)^\circ$ and 70°. Work out the size of the largest angle.

Step 1	Angles in a quadrilateral add to 360°	$x + 10 + 2x + 3x - 20 + 70 = 360$
Step 2	Simplify and solve	$6x + 60 = 360,\ \ x = 50$
Step 3	Work out each angle	$60^\circ, 100^\circ, 130^\circ, 70^\circ$
Step 4	Answer	130°

Practice: angles in a quadrilateral

3.7 The angles of a quadrilateral are $3x^\circ$, $(2x + 10)^\circ$, $(x + 20)^\circ$ and 90°. Work out the size of the largest angle.

3.8 The angles of a quadrilateral are x°, $(x + 30)^\circ$, $2x^\circ$ and $(2x - 30)^\circ$. Work out the size of the largest angle.

3.9 The angles of a quadrilateral are $4x^\circ$, $(3x + 5)^\circ$, $(2x + 15)^\circ$ and 52°. Work out the size of the largest angle.

3.10 The angles of a quadrilateral are $(5x - 10)^\circ$, $(2x + 40)^\circ$, $3x^\circ$ and 80°. Work out the size of the largest angle.

3.11 A kite has angles $(2x + 15)^\circ$, $(2x + 15)^\circ$, $(3x - 10)^\circ$ and $(3x - 10)^\circ$. Work out the size of the largest angle.

3.12 The angles of a quadrilateral are $(x + 25)^\circ$, $(2x - 5)^\circ$, $(2x + 20)^\circ$ and 100°. Work out the size of the largest angle.